The distributive law

1 The student should multiply the second part of the sum, 2, by the number outside the brackets, 6 to get 12 as the second term on the right hand side.

2

a $5s+35$ b $45+9w$

c $3r-18$ d $7y+56$

e $-8t-80$ f $-8c+40$

g $-y-2$ h $16s+20$

i $-80-30r$ j $16t+24$

k $pq+4p$ l $9u-9v$

m $-2m+3n$ n $-11rs-13rt$

o $8r+12s+20t$ p $24m+18n+30p$

3

a $12mn+18m$ b $9v^2+54$

c r^2+7r d $24m^2+42m$

e $-12a^2-54a$ f $4wy^2+4w^2y$

g $16pq-12p+24pr$ h $2f^2-14fg-12fh$

i $-10y^2-70$ j $-9+s^3$

k rs^2+8r l $-30-80u^3$

m $16s^5t^4-36s^5$ n $12p^3q^5-54p^3$

o $9f^2-36fg-54fh$ p $3uv^2+3u^2v$

4

a $4x+30$ b $15x+27$ c $3x+17$ d $18x+32$

e $36x-11$ f $-11x+24$ g $-x-15$ h $-40x+47$

i $10y-49$ j $20x-52$

5

a $x^2+12x+72$ b $15y^2-39y$ c $10x-33y$ d $12y+93$

e $13x-16$ f $4y+44$ g $-11x-26$ h $17a+20$

i $14u^2+20uv$ j $62w^2+34wz$

Applications

7 $(16r - 24)$ representatives

8 $(10 + 15n^4)$ boxes

9 $(16s^2t + 16st^2)$ units^2

10 $(u^2 + 8u)$ m^2

Factorisation

11

a Yes	b No	c Yes	d Yes
e No	f Yes		

12

a 2	b 3	c 3	d 2
e 7	f $4x$	g 5	h 6
i 3			

13

a 2	b 5	c 3	d 2
e $v + 3$	f $a - 4$	g $8a + 3$	h $2w - 9$
i 4	j 5	k 6	l $r + 7$
m $t - 2$	n $9u + 2$		

14

a $6(v + 5)$	b $6(a - 8)$	c $5(9t - 8)$	d $2(m + 5n)$
e $6(9 - 10fg)$	f $8(g - 9h)$	g $3(3f + 5g)$	h $3(w + 5)$
i $2(c - 9)$	j $5(8c - 7)$	k $6(w + 5y)$	l $2(7y + 2w)$
m $2(2a - 7b)$	n $4(8 - 9jk)$	o $2(4jkt - 9)$	p $f(3g - 10h)$
q $rw(3t - 8)$	r $2rt(2 - 3r)$	s $4wy(4 - 9w)$	

15 $2w - 10z + 8y = 2(w - 5z + 4y)$

16

a $4(2g+7h-5)$

b $4(w+z+y)$

c $4(u-3v+5w)$

d $5(2m-5n-3p)$

e $3(m-2n+4p)$

f $5(2g-7h+3)$

g $z(w-y-k)$

h $5r(12-5t-7rt)$

i $4m(3-8n-5mn)$

j $4z\left(4-8y-6y^2\right)$

k $5rt(5r-3t-9rt)$

Factorisation of negatives

17

a $-(4r+3s)$

b $-7(r+s)$

c $-(2+9r)$

d $-(8x+5y)$

18

a $-8(b+2)$

b $-2(9a-8)$

c $-10(8v+3)$

d $-6(6+x)$

e $-5(5u+7v)$

f $-7(5m-7n)$

g $-2pr(4q+3)$

h $-w^2(8-3y)$

i $-3j^2(8+9k)$

j $-m(8m-9n)$

k $-uv(10u-9v)$

l $-3j^2(7+12k)$

m $-2(s+5)$

n $-8\left(v^2-7\right)$

o $-2(6s-5)$

p $-7(4b+3)$

q $-7\left(9r^2-4\right)$

r $-4(m+8n)$

s $-7(7y+3w)$

19

a $-9(2p+q+5r)$

b $-5(2r+5s-4t)$

c $-5(4u+5v-9w)$

d $-2(4w+9y+5z)$